B.Sc. (Part-I) First Semester (NEP) Winter 2024

129200 - Physics DSC - Gravitation & Rotational Dynamics

P. Pages : 2

Time : Two Hours

AG - 7114

Max. Marks : 30

Notes : 1. Attempt all questions.
2. All questions carry equal marks.

1. Multiple Choice Questions (**any six**) 6

i) The relation between gravitational potential and gravitational field intensity is

a) $V = \frac{\partial E}{\partial r}$ b) $E = \frac{\partial V}{\partial r}$

c) $V = -\frac{\partial E}{\partial r}$ d) $E = -\frac{\partial V}{\partial r}$

ii) Relation between acceleration due to gravity and universal gravitational constant G is ---------

a) $GM = gR$ b) $GM = gR^2$

c) $Gg = MR$ d) $Gg = MR^2$

iii) The law of ------ states that areal velocity of a planet remains constant

a) Elliptic orbit b) Areas

c) Periodic time d) Gravitation

iv) S.I. unit of moment of inertia is

a) gm cm b) $gm\,cm^2$

c) kg m d) $kg\,m^2$

v) The moment of linear momentum is ----------

a) linear momentum b) angular momentum

c) torque d) Impulse

vi) M.I. of a uniform disc about a perpendicular axis, passing through its centre is -----

a) MR^2 b) $\frac{1}{2}MR^2$

c) $\frac{1}{3}MR^2$ d) $\frac{3}{2}MR^2$

2. A) Short answer questions (**any one**) 2

a) Define gravitational field intensity and gravitational potential.

b) Write SI unit and dimensions for universal gravitational constant

B) Long answer questions (**any one**) 4

P.T.O

a) Derive an expression of gravitational potential due to uniform spherical shell at a point outside the shell.

b) State and explain Newton's universal law of gravitation.

3. A) Short answer questions (**any one**). 2

a) What is escape velocity?

b) What is the weightlessness condition in a satellite?

B) Long answer questions (**any one**) 4.

a) State Kepler's laws of planetary motion.

b) Derive an expression for variation of acceleration due to gravity with altitude.

4. A) Short answer questions (**any one**) 2

a) What is rigid body? Give the examples.

b) What is torque? State its SI unit.

B) Long answer questions (**any one**) 4

a) State and explain law of conservation of angular momentum.

b) Calculate the rotational kinetic energy and angular momentum of a solid sphere rotating with angular velocity 10 rad/sec. (Given M.I. of solid sphere about its diameter $= 0.008\,kg.m^2$.)

5. A) Short answer questions (**any one**) 2

a) M.I. of a hollow sphere about an axis passing through its diameter is $\frac{2}{3}MR^2$, find the MI about a tangent.

b) State law of parallel axes of moment of inertia.

B) Long answer questions (**any one**) 4

a) Deduce the formula for moment of inertia of a circular ring about an axis passing through its centre and perpendicular to its plane.

b) State and explain law of perpendicular axes of moment of inertia.

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